

Week 3 - Clustering & Classification

1. Introduction

This document provides a detailed explanation of the 'week3_clustering&classification' notebook. It covers two major parts: Clustering (unsupervised learning) using K-Means and Classification (supervised learning) using Random Forest. The dataset used is the Netflix dataset, containing information about Movies and TV Shows along with their genres, duration, rating, and other features.

2. Data Loading & Overview

The dataset 'netflix_edu.csv' is loaded using pandas. It contains columns such as show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, description, num_seasons, and rating_group. The goal is to use these features for clustering and classification tasks.

3. K-Means Clustering

K-Means is an unsupervised learning algorithm used to group similar data points into K clusters. In this notebook, K-Means was used to group Netflix titles based on genre, duration, and rating.

Steps followed:

- 1 Genre features were extracted using MultiLabelBinarizer on the 'listed_in' column.
- 2 Duration was converted into numeric minutes, treating each TV season as approximately 60 minutes.
- 3 Rating groups were mapped into numeric codes using a dictionary mapping.
- 4 Features were standardized using StandardScaler to bring all values onto the same scale.
- 5 K-Means with 5 clusters was applied to the standardized data.
- 6 Cluster labels were assigned back to the dataframe for analysis.
- 7 Cluster counts were calculated and visualized.

4. PCA Visualization of Clusters

Principal Component Analysis (PCA) was applied to reduce high-dimensional feature data into 2 dimensions for easy visualization. The two principal components (PCA1 and PCA2) were plotted using seaborn, where each cluster is represented in a distinct color. Cluster centroids were also plotted as 'X' markers.

5. Classification using Random Forest

The goal of the classification part was to predict whether a given content is a Movie or a TV Show. Random Forest, a powerful ensemble learning algorithm, was used for this purpose.

Steps followed:

- 1 A new column 'primary_genre' was created by extracting the first genre from 'listed_in'.
- 2 'content_age_years' was computed as the difference between the current year and the release year.
- 3 'rating_group' and 'primary_genre' were encoded into numeric category codes.
- 4 Features such as duration, rating_group, content_age_years, and primary_genre were selected for X.
- 5 The target y was encoded as 0 for Movie and 1 for TV Show.
- 6 A Random Forest Classifier was trained and tested on this dataset.
- 7 Model performance was evaluated using a Confusion Matrix and Accuracy Score.

6. Feature Importance

Feature importance was visualized using a horizontal bar plot, where each bar represented how much a feature contributed to the classification. Features such as 'duration' and 'content_age_years' were often more important in differentiating between Movies and TV Shows.

7. Confusion Matrix Visualization

A confusion matrix was plotted using seaborn heatmap to display the number of correct and incorrect predictions. Different color palettes (like 'Blues', 'OrRd', 'plasma', etc.) were explored to enhance visual understanding.

8. Conclusion

The analysis successfully demonstrated the use of unsupervised learning (K-Means Clustering) and supervised learning (Random Forest Classification) on the Netflix dataset. Clustering helped identify patterns in genres and durations, while classification accurately differentiated between Movies and TV Shows. The combination of PCA visualization and feature importance interpretation provided deeper insight into the data.